

Contact

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(LinkedIn)
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Top Skills

Amazon Web Services (AWS)
AWS Command Line Interface (CLI)
AWS Lambda

Certifications

Introduction to Complex Analysis
Principles of Computing (Part 1)
Structuring Machine Learning
Projects
Sequence Models
The Fundamentals of Computing
Capstone Exam

James Batman

Software Engineer — Cloud Infrastructure, Data Pipelines & IoT
Systems
Portland, Oregon Metropolitan Area

Summary

After 17 years as a commodity trader managing risk and operational complexity under market pressure, I made a deliberate move into software engineering. My focus is cloud infrastructure and data pipelines on AWS — IoT device integration, Lambda-based architectures, and infrastructure as code — now working primarily in Python and Go, with an increasing investment in AI-assisted development tooling. Self-directed by nature, I care about understanding systems deeply, not just making them work, and have built enough range across domains to be regularly trusted with projects well outside any single specialty.

Experience

UBX Systems Inc.

Software Engineer

September 2024 - Present (1 year 10 months)

Parkdale, Oregon

IoT Device Integration & Onboarding

- Onboarded and integrated diverse LoRaWAN sensors (thermostats, current transformers, occupancy counters, leak detection, CO₂, smart buttons) including packet decoding, Lambda development, and AWS IoT provisioning
- Developed a unified decoder architecture; translated vendor JavaScript decoders to Python with LLM-assisted tooling
- Scripted device configuration-by-downlink workflows for remote device management at scale

AWS Lambda & Cloud Pipeline Development

- Built and maintained Lambda functions for IoT data ingestion, decoding, routing, and enrichment; refactored the primary ingest Lambda with a rollback deployment strategy
- Implemented cross-account MQTT routing via IAM roles and SQS for external client data delivery
- Built a token-rotation Lambda using VPC + NAT gateway with secrets in AWS Secrets Manager

Firmware Over-the-Air (FUOTA)

- Scripted automated FUOTA workflows including IAM configuration, S3 artifact management, and AWS IoT task orchestration

Operational Tooling & Monitoring

- Built a daily device-health Lambda querying time-series data for reporting gaps and low battery, with EventBridge cron and Slack alerts
- CLI diagnostic tools surfacing device connectivity issues, gateway outages, and LTE signal strength

Data Engineering

- Wrote complex SQL queries for energy and time-series data; built automated S3-to-database ingestion pipelines
- Delivered on-demand multi-site sensor data exports; identified and corrected a scaling bug in accumulated energy data

Frontend Development (Vue.js)

- Built energy reporting UI components (bar charts, building comparisons, IAQ reports) using Vue.js and charting libraries; contributed GraphQL query development for energy, occupancy, and IAQ endpoints

Image Processing

- EventBridge + Lambda pipeline for S3 thumbnail generation supporting JPEG, PNG, HEIC/HEIF, and MPO; compiled custom ARM64 Lambda layers (Pillow + libheif)

Ventacity Systems

Software Engineer

March 2022 - December 2024 (2 years 10 months)

Portland, Oregon, United States

self-employed

Student

October 2017 - June 2021 (3 years 9 months)

Portland, OR

Self-employed

Commodity Trader

January 2000 - June 2017 (17 years 6 months)

Education

The College of Wooster

Bachelor of Arts - BA, Chemistry; Mathematics minor

